

Unless otherwise indicated, each question is of equal value [4 pts]. Show all of your work.

Use $\mathbf{u} = \langle 2, 4, 1 \rangle$ and $\mathbf{v} = \langle 1, -3, 2 \rangle$ to perform the indicated operations in problems 1–3.

1. $\mathbf{u} - 2\mathbf{v}$

$$= \langle 2 - 2, 4 + 6, 1 - 4 \rangle = \langle 0, 10, -3 \rangle$$

2. $\mathbf{u} \cdot \mathbf{v}$

$$= 2(1) + 4(-3) + 1(2) = 2 - 12 + 2 = -8$$

3. $\mathbf{u} \times \mathbf{v}$

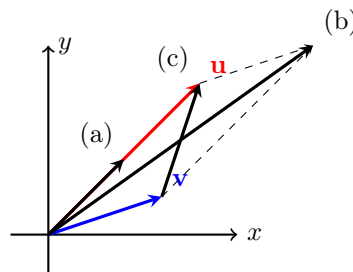
$$= \det \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 4 & 1 \\ 1 & -3 & 2 \end{vmatrix} = \det \begin{vmatrix} 4 & 1 \\ -3 & 2 \end{vmatrix} \mathbf{i} - \det \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} \mathbf{j} + \det \begin{vmatrix} 2 & 4 \\ 1 & -3 \end{vmatrix} \mathbf{k} = 5\mathbf{i} - 3\mathbf{j} - 10\mathbf{k} \text{ or } \langle 5, -3, -10 \rangle$$

Use the graph to draw the following vectors in problems 4–5 (provide labels). Make certain to indicate direction.

4. a. $\frac{1}{2}\mathbf{u}$

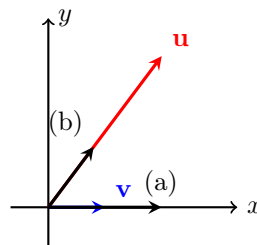
b. $\mathbf{u} + \mathbf{v}$

c. $\mathbf{u} - \mathbf{v}$ (you do not have to draw it as a position vector)



5. a. $\text{proj}_{\mathbf{v}} \mathbf{u}$

b. $\text{proj}_{\mathbf{u}} \mathbf{v}$



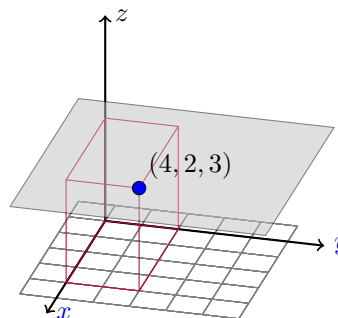
Use the graph to perform the indicated operations (provide labels).

6. a. Use the right-hand rule to label the x and y axes.

Shown in blue.

b. Plot the point $(4, 2, 3)$.

c. Sketch the plane $(x, y, 3)$.



Use the vectors $\mathbf{u} = \langle 3, 6 \rangle$, $\mathbf{v} = \langle 2, 3 \rangle$ and the point $P(1, -4)$ to complete problems 7–10.

7. Find the line through P and parallel to $\mathbf{u}$.

$$L = (1, -4) + t(3, 6) \text{ (answers may vary)}$$

8. Find the angle between $\mathbf{u}$ and $\mathbf{v}$.

$$\mathbf{u} \cdot \mathbf{v} = 3(2) + 6(3) = 24, \theta = \cos^{-1} \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} \right) = \cos^{-1} \left(\frac{24}{\sqrt{45}\sqrt{13}} \right) \approx .1244 \text{ radians (or } \approx 7.125^\circ \text{)}.$$

9. Find a vector orthogonal to $\mathbf{u}$ and $\mathbf{v}$.

This question was written for vectors in $\mathbb{R}^3$. I made them 2d to make the computations above easier, however I should have edited this problem. There is no vector perpendicular to both of these vectors as their angle is not 0 or $\pm\pi$. Sorry for the mistake I am excluding this problem. For 3d vectors an orthogonal vector can be found using the cross-product.

10. Is the vector found in problem 9 unique? If not find another vector orthogonal to $\mathbf{u}$ and $\mathbf{v}$.

As stated above, the vector does not exist. For non-parallel 3d vectors, an orthogonal vector always exists and is NOT unique. Another orthogonal vector can be found by scaling.

Let $\mathbf{r}(t) = \langle 1, 2t^{-2}, e^{-t} \rangle$. Evaluate problems 11–13.

11. $\mathbf{r}'(t)$

$$\mathbf{r}'(t) = \langle 0, -4t^{-3}, -e^{-t} \rangle$$

12. $\int_0^1 \mathbf{r}(t) dt$

$$\int_0^1 \mathbf{r}(t) dt = \langle t, -2t^{-1}, -e^{-t} \rangle \Big|_0^1 = \langle 1 - 0, -2 - 0, -e^{-1} - 0 \rangle = \langle 1, -2, -e^{-1} \rangle$$

13. $\lim_{t \rightarrow \infty} \mathbf{r}(t)$

$$\lim_{t \rightarrow \infty} \mathbf{r}(t) = \left\langle \lim_{t \rightarrow \infty} 1, \lim_{t \rightarrow \infty} 2t^{-2}, \lim_{t \rightarrow \infty} e^{-t} \right\rangle = \langle 1, 0, 0 \rangle$$

Compute the slope of the tangent line of the vector valued function at $t = 1$.

14. $\mathbf{r}(t) = \langle 1 + t^3, 2 \sin 3t, e^t \rangle$

$$\mathbf{r}'(t) = \langle 3t^2, 6 \cos 3t, e^t \rangle$$

$$\mathbf{r}'(1) = \langle 3, 6 \cos(3), e \rangle$$

Complete problems 15–17.

15. A sailboat floats in a current that flows due west at 3 m/s. The wind is blowing 30 north of east at a speed of 5 m/s. Find the speed of the boat relative to the shore.

a. $\mathbf{s} = \langle -3, 0 \rangle + \langle 5 \cos 30^\circ, 5 \sin 30^\circ \rangle = \left\langle -3 + \frac{5\sqrt{3}}{2}, \frac{5}{2} \right\rangle$

$$|\mathbf{s}| = \sqrt{\left(-3 + \frac{5\sqrt{3}}{2}\right)^2 + \left(\frac{5}{2}\right)^2} \approx 2.832 \text{ m/s}$$

16. A constant force vector $\mathbf{F} = \langle 50, 120 \rangle$ (pounds) is used to move a heavy object 50 feet along the x -axis. Calculate the work done.

$\mathbf{d} = \langle 50, 0 \rangle$, so $\mathbf{F} \cdot \mathbf{d} = 50^2 = 2500 \text{ N}$

17. A 40-pound weight hangs from the end of a 2-foot bar that is attached to a wall at an angle of 45° below the horizontal.

- b. Find the magnitude of the torque about the point on the wall where the bar is attached.

$$|\tau| = 2(40) \sin 45^\circ = 80 \frac{\sqrt{2}}{2} \approx 56.568$$

- c. Draw a picture and indicate the direction of the torque.

